Ch-10 Simple linear regression - (10M)

* It is a statical method used to model the relationship between one dependent variable at a single independent variable.
* The simple linear regression line equation on y on x
* It is generally Denoted by the relation $y = a$ where a, b are called regression cofficient, x is independent variable and y is dependent variable is called simple linear regression line equation y on x.
* where $a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n\Sigma x^2 - (\Sigma x)^2}$

$$b = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

* The Simple linear regression line equation on x on y
* let the line equation of the regression line be $x = c + dy$

where c, d are called regression coefficients. y is Dependent Variable and y is independent variable.

$$c = \frac{(\Sigma x)(\Sigma y^2) - (\Sigma y)(\Sigma xy)}{n\Sigma y^2 - (\Sigma y)^2}$$

$$d = \frac{n\Sigma xy - (\Sigma y)(\Sigma x)}{n\Sigma y^2 - (\Sigma y)^2}$$

Find the Simple linear regression coefficients the following data. using x as independent and y is Dependent variable.

x	1	3	4	6	8	9	11	14
y	1	2	2	4	5	7	8	10

let the eq^n of the regression line be

$y = a + bx \longrightarrow (1)$

x	y	x^2	xy
1	1	1	1
3	2	9	6
4	2	16	8
6	4	36	24
8	5	64	40
9	7	81	63
11	8	121	88
14	10	196	140
$\Sigma x = 56$	$\Sigma y = 39$	$\Sigma x^2 = 524$	$\Sigma xy = 370$

...heorem.
...n it was selected? Bayes' theorem is used
...ayes was a British mathematician who was given the above theorem
...practical problems

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n\,\Sigma x^2 - (\Sigma x)^2}$$

$$a = \frac{(39)(524) - (56)(370)}{8[524) - (56)^2}$$

$$a = \frac{20436 - 20720}{4192 - 3136}$$

$$a = \frac{-284}{1056}$$

$$a = -0.26$$

$$b = \frac{n\,\Sigma xy - (\Sigma x)(\Sigma y)}{n\,\Sigma x^2 - (\Sigma x)^2}$$

$$b = \frac{8(370) - (56)(39)}{8(524) - (56)^2}$$

$$b = \frac{2960 - 2184}{4192 - 3136}$$

$$b = \frac{776}{1056}$$

$$b = 0.73$$

Substitube a, b values in eq (1)

$$y = -0.26 + 0.73x$$

$$c = \frac{(\Sigma x)(\Sigma y) - (\Sigma y)(\Sigma xy)}{n\Sigma y^2 - (\Sigma yx)}$$

$$c = \frac{(56)(263) - (39)(370)}{8(263) - (39)^2}$$

$$c = \frac{14728 - 14430}{2104 - 1521}$$

$$c = \frac{298}{583}$$

$$c = 0.51$$

$$d = \frac{n\Sigma xy - (\Sigma y)(\Sigma x)}{n\Sigma y^2 - (\Sigma y)^2}$$

$$d = \frac{8(370) - (39)(56)}{8(263) - (39)^2}$$

$$d = \frac{2960 - 2184}{2104 - 1521}$$

$$d = \frac{776}{583}$$

$$d = 1.33$$

was selected? Bayes' theorem is used

was a British mathematician who was given the above theorem

Find the simple linear regression cofficients to the following data.

x	4	2	8	6
y	7	3	10	5

using x as independent variable and y as dependent variable.

let the eqn of regression line will be $y = a + bx \longrightarrow (1)$

x	y	x^2	xy
4	7	16	28
2	3	4	6
8	10	64	80
6	5	36	30

$\boxed{n = 4}$

$\Sigma x = 20 \quad \Sigma y = 25 \quad \Sigma x^2 = 120 \quad \Sigma xy = 144$

$$a = \frac{\Sigma y \, \Sigma x^2 - \Sigma x \, \Sigma xy}{n \Sigma x^2 - (\Sigma x)^2}$$

$$a = \frac{25(120) - 20(144)}{4(120) - (20)^2}$$

$$a = \frac{3000 - 2880}{480 - 400}$$

$$a = \frac{120}{80}$$

$$\boxed{a = 1.5}$$

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$b = \frac{4(144) - 20(25)}{4(120) - (20)^2}$$

$$b = \frac{576 - 500}{480 - 400}$$

$$b = \frac{76}{80}$$

$$\boxed{b = 0.95}$$

Substitube a, b values in eq(1)

$$y = 1.5 + 0.95x$$